GRADED LAB EXAMINATION

Course: Operating Systems E-9 **Total Marks:** 30

General Instruction:

- Attempt all questions.
- Use appropriate commands and programming logic.
- Ensure that your programs produce the required output.
- Marks will be awarded based on the correctness of solution, completion of task, and performance in the viva
- Submissions delayed by up to 3 minutes will incur a **25% deduction** from the total marks. Submissions delayed by more than 3 minutes will incur a **50% deduction** from the total marks.

Question 1: Perform the following tasks using operating system commands. **(10 marks)**

1. Create a directory named Graded Lab in your home directory. Inside the Graded Lab, create two subdirectories named Student and Teacher.
2. Create a file named student.txt inside the student directory and write the following sentences in it:

   ```
   I am a BSCS student.
   I am learning operating system commands.
   ```

3. Create another file named teacher.txt inside the Teacher directory and write following sentences in it:

   ```
   I am a teacher.
   I teach Linux commands.
   ```

4. Display the contents of both student.txt and teacher.txt using appropriate Linux commands.
5. Rename student.txt using your own name, for example, awais.txt. Keep the renamed file inside the student directory.

Question 2: Write a C program that creates an integer array of size 10 and accepts 10 integers from the user. Add all even numbers and subtract all odd numbers to calculate the final result. **(10 marks)**

Example Input: 2 5 4 7 6 3 8 1 10 9

Calculation: 2 - 5 + 4 - 7 + 6 - 3 + 8 - 1 + 10 - 9

Expected Output: Final Result: 5

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Question 3: Write a C program that accepts integers through command-line arguments and performs the following tasks: **(10 marks)**

1. Count and display the total number of even and odd integers provided.
2. Reverse the digits of each integer individually.
3. If a reversed number is even, add its corresponding argument index to that number. Otherwise, subtract the argument index from that number.
4. Calculate and display the final sum of all modified numbers.

Instructions:

- Use argc and argv to process command-line arguments.
- Convert the arguments into integers before performing calculations.
- Assume that at least one integer is provided.
- Clearly display the even count, odd count, and final sum.